

Experiment 5

1. Title: Apply array methods and object handling; create a cart total calculator with discount logic.

2. Software/Tools Required:

- Visual Studio Code (or any text editor)
- Web Browser (Google Chrome/Microsoft Edge/Mozilla Firefox)

3. Theory:

Array Methods:

Array methods are built-in JavaScript functions used to perform operations on arrays efficiently. Common methods include `push()`, `pop()`, `map()`, `filter()`, `reduce()`, `find()`, and `forEach()`. They are useful for adding, removing, searching, filtering, transforming, and processing array elements.

Object Handling:

Objects in JavaScript are collections of **key-value pairs** used to represent structured data. Object properties can be accessed, modified, added, or removed as required. Objects are commonly used to represent entities such as products, users, or orders.

Array methods help process collections of data, while objects organize related data into meaningful structures. Together, they provide an efficient way to implement applications such as shopping carts and order systems.

4. Output:

Shopping Cart Calculator

SmartPhone 75000 2 Add Product

ID	Product	Price	Quantity	Total
1	Laptop	40000	1	40000
2	SmartPhone	75000	2	150000

Total Amount : 190000
Discount : 38000.00
Final Amount : 152000.00

C:/Users/karti/OneDrive/Desktop/AI_ML_Projects/JS_Lab_6/index.html

Maximum & Minimum Finder

Enter numbers one by one and click **Add Number** (or press Enter). When done, click **Submit**.

Enter a number Add Number

Entered Numbers:
5, 12, 4, 11, 25

Submit

Minimum Number
4

Maximum Number
25

10:43
10-08-2026

Case Study : *Minimum and Maximum Number Identification from Array*

The program accepts numbers from the user and stores them in an array after validating the input. It uses array operations to add and display the entered numbers dynamically. A separate function traverses the array to find the minimum and maximum values by comparing each element. The calculated minimum and maximum values are then displayed on the webpage using DOM manipulation. The program also supports adding numbers by pressing the Enter key.

5. Result/Conclusion:

The experiment successfully demonstrated the use of **JavaScript arrays, objects, and array methods** such as `map()`, `reduce()`, and `forEach()`. A cart total calculator was implemented to calculate item-wise prices, subtotal, apply **conditional discount logic**, and determine the final payable amount. This experiment helped understand how JavaScript can efficiently handle and process structured data using arrays and objects.